

GED Science Practice Test 6

PART A — QUESTIONS

GED-SCI6-001

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student studies how pH affects the activity of catalase, an enzyme found in many living cells. Catalase breaks down hydrogen peroxide into water and oxygen. Four test tubes contain the same amount of catalase and hydrogen peroxide, but each tube has a different pH. The student measures the oxygen produced in 2 minutes.

pH: 3, 5, 7, 9

Oxygen produced (mL): 4, 11, 28, 16

Which conclusion is best supported by the data?

- A) Catalase was most active at pH 7.
- B) Catalase was most active at pH 3.
- C) pH had no effect on catalase activity.
- D) Catalase produced the same oxygen amount at every pH.

GED-SCI6-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

Pressure can be calculated as force divided by area. A student pushes four objects with the same force, but each object touches the floor over a different area.

Object | Force (N) | Contact area (m²)

A | 120 | 2

B | 120 | 3

C | 120 | 4

D | 120 | 6

Which object applies the greatest pressure to the floor?

- A) Object A
- B) Object B
- C) Object C
- D) Object D

GED-SCI6-003

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

Astronomers classify stars partly by color and temperature. Blue-white stars are generally hotter than yellow stars. Red stars are generally cooler than yellow stars. The Sun is a yellow star with a surface temperature of about 5,800 K.

Which star would most likely be hotter than the Sun?

- A) A blue-white star
- B) A red star
- C) A small dark asteroid
- D) A rocky planet

GED-SCI6-004

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The liver performs many functions that help maintain a stable internal environment. It processes nutrients absorbed from the digestive system, stores some glucose as glycogen, breaks down certain toxins, and produces bile that helps digest fats.

Which function of the liver is described as helping remove harmful substances from the body?

- A) Breaking down certain toxins
- B) Moving air into the lungs
- C) Sending nerve signals to muscles
- D) Forming bones from calcium

GED-SCI6-005

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A box applies a force of 600 N over an area of 3 m².

Pressure = force ÷ area

What pressure does the box apply in N/m²?

Type your answer as a whole number.

GED-SCI6-006

Section: Science

Type: multi_select

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student investigates how the position of a fulcrum affects the force needed to lift a load with a lever. The student changes the fulcrum position but uses the same load and the same lever in each trial.

Which TWO factors should be kept constant?

- A) Mass of the load
- B) Length of the lever
- C) Position of the fulcrum
- D) Force needed to lift the load
- E) Final measured result

GED-SCI6-007

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

In some flowers, crossing a red-flowered plant with a white-flowered plant produces pink-flowered offspring. This pattern is called [D1] because the heterozygous offspring show [D2].

Dropdowns:

D1) incomplete dominance | active transport | succession

D2) an intermediate trait | no inherited traits | only the recessive trait

GED-SCI6-008

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student tests a clear liquid with red and blue litmus paper. Blue litmus paper turns red, while red litmus paper stays red. The liquid also reacts with magnesium metal and produces bubbles of gas.

Which type of substance is most likely being tested?

- A) Acid
- B) Neutral salt
- C) Pure water
- D) Inert gas

GED-SCI6-009

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A community compares energy sources. Petroleum forms from ancient organisms over millions of years and is used faster than it forms. Geothermal energy comes from heat inside Earth and can be used repeatedly in certain areas.

Which statement best identifies the nonrenewable resource?

- A) Petroleum is nonrenewable because it forms too slowly to be replaced on a human time scale.
- B) Geothermal energy is nonrenewable because Earth has no internal heat.
- C) Petroleum is renewable because it can be pumped from underground.
- D) Geothermal energy is renewable only if it is burned.

GED-SCI6-010

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A wildlife biologist studies the population of deer in a fenced preserve. The preserve has limited food, water, and space.

Year | Deer population

1 | 45

2 | 81

3 | 126

4 | 149

5 | 153

6 | 151

Which conclusion is best supported by the data?

- A) The deer population appears to level off near the preserve's carrying capacity.
- B) The deer population increases by the same number every year.
- C) The deer population reaches zero by Year 6.
- D) The preserve has unlimited resources for deer.

GED-SCI6-011

Section: Science

Type: drag_drop

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Match each target with the correct tile.

Tiles:

- A) Type of insulation wrapped around each cup
- B) Temperature change of the water after 20 minutes
- C) Same starting water temperature in every cup
- D) Brand name printed on the cup

Targets:

- T1) Independent variable
- T2) Dependent variable
- T3) Controlled variable

GED-SCI6-012

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A person bends an arm at the elbow to lift a bag. The biceps muscle contracts while the triceps muscle relaxes. To straighten the arm again, the triceps contracts while the biceps relaxes.

Which statement best describes how these muscles work?

- A) They work as an opposing pair to move the arm in different directions.

- B) They both contract at the same time to prevent movement.
- C) They digest food before movement can happen.
- D) They carry oxygen through blood vessels.

GED-SCI6-013

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A cold pack contains two substances in separate compartments. When the inner compartment is broken, the substances mix and the pack becomes cold. The reaction absorbs thermal energy from the surroundings.

Which statement best describes this reaction?

- A) It is endothermic because it absorbs thermal energy.
- B) It is exothermic because it absorbs thermal energy.
- C) It creates matter from nothing.
- D) It stops all particle motion immediately.

GED-SCI6-014

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Atmospheric CO₂ in one location increased from 400 ppm to 420 ppm over a period of years.

What was the percent increase?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI6-015

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A researcher records urine output for a healthy adult during two conditions.

Condition	Water intake (L/day)	Urine output (L/day)
Low water intake	1.0	0.6
High water intake	2.5	1.9

Which conclusion is best supported by the data?

- A) Urine output increased when water intake increased.
- B) Urine output stayed the same in both conditions.
- C) The kidneys stopped working during high water intake.
- D) Higher water intake caused urine output to decrease to zero.

GED-SCI6-016

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student compares how well different wall materials reduce sound. The student places each material between a speaker and a sound meter. The speaker volume, speaker distance, and sound-meter position stay the same. The student records the sound level after the sound passes through each material.

What is the dependent variable?

- A) Sound level recorded by the meter
- B) Type of wall material
- C) Speaker volume
- D) Distance from speaker to meter

GED-SCI6-017

Section: Science

Type: dropdown

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A snowshoe spreads a person's weight over a larger area than a regular boot. For the same weight, increasing the contact area [D1] the pressure on the snow, so the person is [D2] likely to sink deeply.

Dropdowns:

- D1) decreases | increases | eliminates
- D2) less | more | equally

GED-SCI6-018

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies how exercise intensity affects lactic acid levels in muscle cells. Volunteers complete light, moderate, and intense cycling sessions for the same amount of time. The researcher measures blood lactate immediately after each session.

Which factor is the independent variable?

- A) Exercise intensity
- B) Blood lactate level
- C) Cycling time
- D) Type of measurement device

GED-SCI6-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student builds circuits using one resistor and changes the voltage. The current is recorded.

Voltage (V): 2, 4, 6, 8

Current (A): 0.5, 1.0, 1.5, 2.0

Which statement best describes the relationship in the data?

- A) Current increases in direct proportion to voltage.
- B) Current decreases as voltage increases.
- C) Current stays the same as voltage changes.
- D) Current becomes zero when voltage reaches 8 V.

GED-SCI6-020

Section: Science

Type: multi_select

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A soil scientist compares two soil samples.

Soil sample | Sand (%) | Clay (%) | Water infiltration after 10 min (mL)

Sample A | 70 | 10 | 85

Sample B | 25 | 45 | 32

Which TWO statements are best supported by the data?

- A) Sample A allowed more water to infiltrate than Sample B.
- B) Sample B had a higher clay percentage than Sample A.
- C) Sample B allowed more water to infiltrate than Sample A.
- D) Sample A and Sample B had the same clay percentage.
- E) Sample A had less sand than Sample B.

GED-SCI6-021

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Human blood type can be affected by multiple alleles. A person with type AB blood has one A allele and one B allele. Both alleles are expressed in the blood type instead of one completely hiding the other.

Which inheritance pattern is described?

- A) Codominance
- B) Photosynthesis
- C) Decomposition
- D) Evaporation

GED-SCI6-022

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student uses a lever to lift a load. Mechanical advantage can be calculated as output force divided by input force.

Trial | Output force (N) | Input force (N)

A | 120 | 60

B | 120 | 40

C | 120 | 30

D | 120 | 20

Which trial has the greatest mechanical advantage?

A) Trial A

B) Trial B

C) Trial C

D) Trial D

GED-SCI6-023

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A text model describes the life cycle of a medium-sized star. A cloud of gas and dust collapses under gravity, forming a protostar. The star then spends most of its life fusing hydrogen in its core. Later, it expands into a red giant and eventually leaves behind a dense white dwarf.

According to the model, which stage occurs before the star becomes a main-sequence star?

A) Protostar

B) White dwarf

C) Red giant

D) Black hole

GED-SCI6-024

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A clinic tests 250 blood samples. Of these samples, 35 have low iron levels.

What percent of the samples have low iron levels?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI6-025

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A mixture contains sand, salt, and iron filings. A student first uses a magnet to remove the iron filings. Then the student adds water, filters out the sand, and evaporates the water to recover the salt.

Which property allows the iron filings to be separated first?

- A) Magnetism
- B) Boiling point of water
- C) Solubility of salt
- D) Color of sand

GED-SCI6-026

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A cell packages proteins into small sacs that can be moved to other parts of the cell or sent outside the cell. This packaging and shipping function is mainly performed by the Golgi apparatus.

Which statement best describes the Golgi apparatus?

- A) It modifies and packages proteins for transport.
- B) It stores most inherited information in the cell.
- C) It controls voluntary muscle movement.
- D) It absorbs sunlight for photosynthesis in animal cells.

GED-SCI6-027

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student pulls the same block across four surfaces and records the force needed to keep the block moving at a constant speed.

Surface | Pulling force (N)

Waxed wood | 6

Cardboard | 11

Rubber mat | 18

Rough carpet | 25

Which surface produced the greatest friction force?

- A) Waxed wood
- B) Cardboard
- C) Rubber mat
- D) Rough carpet

GED-SCI6-028

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares two ground surfaces on a sunny day.

Surface	Percent sunlight reflected	Afternoon surface temperature (°C)
Light concrete	42	31
Dark asphalt	8	47

Which conclusion is best supported by the data?

A) The darker surface reflected less sunlight and became warmer.

B) The lighter surface reflected less sunlight and became warmer.

C) Both surfaces reflected the same amount of sunlight.

D) Reflection percentage had no relationship to surface temperature.

GED-SCI6-029

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Some fungi grow around plant roots. The fungi help the plant absorb minerals from the soil, and the plant provides the fungi with sugars made during photosynthesis. Both organisms benefit from the relationship.

Which type of relationship is described?

A) Mutualism

B) Predation

C) Parasitism

D) Competition

GED-SCI6-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student tests three household liquids with an indicator that changes color at different pH levels.

Liquid	Indicator color	Approximate pH
Lemon juice	Red	2
Milk	Yellow-green	6
Soap solution	Blue	10

Which liquid is most basic?

A) Lemon juice

B) Milk

C) Soap solution

D) All three liquids have the same pH

GED-SCI6-031

Section: Science

Type: multi_select

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student studies how soil moisture affects the hatching rate of reptile eggs. The student places equal numbers of eggs in containers with low, medium, and high soil moisture. Temperature and light exposure are kept the same.

Which TWO choices would improve the investigation?

- A) Use the same temperature for all containers.
- B) Repeat the test with several containers at each moisture level.
- C) Change both soil moisture and temperature at the same time.
- D) Count only the eggs that hatch first.
- E) Use different egg numbers in each container.

GED-SCI6-032

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A device uses 48 joules of energy in 12 seconds.

Power = energy ÷ time

What is the power of the device in watts?

Type your answer as a whole number.

GED-SCI6-033

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

After a wildfire, grasses begin growing in an open area. Over several years, shrubs appear, followed by young trees. After many more years, the area may develop into a mature forest community if conditions allow.

Which ecological process is described?

- A) Succession
- B) Cellular respiration
- C) Reflex action
- D) Blood clotting

GED-SCI6-034

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A pedigree study tracks a trait in one family. The trait appears in every generation shown. An affected parent and an unaffected parent have four children: two affected and two unaffected. Both males and females can show the trait.

Which inheritance pattern is most consistent with this information?

- A) A dominant trait passed from affected parents to some offspring
- B) A trait caused only by lack of sunlight
- C) A trait that cannot be inherited
- D) A recessive trait that must skip every generation

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI6-001 — Correct: A — Correct Answer: Catalase was most active at pH 7. —

Explanation: The most oxygen, 28 mL, was produced at pH 7, showing the highest catalase activity.

GED-SCI6-002 — Correct: A — Correct Answer: Object A — Explanation: With the same force, the smallest contact area produces the greatest pressure. Object A has the smallest area.

GED-SCI6-005 — Correct: 200 — Tolerance: exact — Correct Answer: 200 — Explanation: $\text{Pressure} = 600 \text{ N} \div 3 \text{ m}^2 = 200 \text{ N/m}^2$.

GED-SCI6-003 — Correct: A — Correct Answer: A blue-white star — Explanation: Blue-white stars are generally hotter than yellow stars such as the Sun.

GED-SCI6-004 — Correct: A — Correct Answer: Breaking down certain toxins — Explanation: Breaking down toxins helps remove or reduce harmful substances in the body.

GED-SCI6-006 — Correct: A,B — Correct Answer: Mass of the load | Length of the lever — Explanation: These factors should stay the same so fulcrum position is the main variable being tested.

GED-SCI6-007 — Correct: D1=incomplete dominance; D2=an intermediate trait — Correct Answer: D1=incomplete dominance; D2=an intermediate trait — Explanation: In incomplete dominance, heterozygous offspring show an intermediate phenotype, such as pink flowers.

GED-SCI6-008 — Correct: A — Correct Answer: Acid — Explanation: Acids turn blue litmus paper red and can react with some metals to produce gas.

GED-SCI6-009 — Correct: A — Correct Answer: Petroleum is nonrenewable because it forms too slowly to be replaced on a human time scale. — Explanation: Petroleum takes millions of years to form and is used much faster than it is naturally replaced.

GED-SCI6-010 — Correct: A — Correct Answer: The deer population appears to level off near the preserve's carrying capacity. — Explanation: The population rises quickly at first and then levels near 150, suggesting a carrying capacity.

GED-SCI6-011 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=Type of insulation wrapped around each cup; T2=Temperature change of the water after 20 minutes; T3=Same starting water temperature in every cup — Explanation: The insulation type is changed, the temperature change is measured, and the starting temperature is controlled.

GED-SCI6-012 — Correct: A — Correct Answer: They work as an opposing pair to move the arm in different directions. — Explanation: The biceps and triceps contract and relax in opposite patterns to bend and straighten the arm.

GED-SCI6-013 — Correct: A — Correct Answer: It is endothermic because it absorbs thermal energy. — Explanation: A reaction that absorbs thermal energy from the surroundings is endothermic.

GED-SCI6-014 — Correct: 5 — Tolerance: exact — Correct Answer: 5 — Explanation: The increase is 20 ppm. Percent increase = $20 \div 400 \times 100 = 5\%$.

GED-SCI6-015 — Correct: A — Correct Answer: Urine output increased when water intake increased. — Explanation: The table shows urine output increasing from 0.6 L/day to 1.9 L/day when water intake increased.

GED-SCI6-016 — Correct: A — Correct Answer: Sound level recorded by the meter — Explanation: The dependent variable is the measured result after changing the wall material.

GED-SCI6-017 — Correct: D1=decreases; D2=less — Correct Answer: D1=decreases; D2=less — Explanation: Spreading the same weight over a larger area decreases pressure, making sinking less likely.

GED-SCI6-018 — Correct: A — Correct Answer: Exercise intensity — Explanation: Exercise intensity is the factor intentionally changed by the researcher.

GED-SCI6-019 — Correct: A — Correct Answer: Current increases in direct proportion to voltage. — Explanation: Current increases by 0.5 A for each 2 V increase, showing a direct proportional relationship.

GED-SCI6-020 — Correct: A,B — Correct Answer: Sample A allowed more water to infiltrate than Sample B. | Sample B had a higher clay percentage than Sample A. — Explanation: Sample A had 85 mL of infiltration compared with 32 mL for Sample B, and Sample B had more clay.

GED-SCI6-021 — Correct: A — Correct Answer: Codominance — Explanation: In codominance, both alleles are expressed, as in type AB blood.

GED-SCI6-022 — Correct: D — Correct Answer: Trial D — Explanation: Mechanical advantage = output force $\div$ input force. Trial D is $120 \div 20 = 6$, the greatest value.

GED-SCI6-023 — Correct: A — Correct Answer: Protostar — Explanation: The model states that a protostar forms before the star spends most of its life fusing hydrogen.

GED-SCI6-024 — Correct: 14 — Tolerance: exact — Correct Answer: 14 — Explanation: $35 \div 250 \times 100 = 14\%$.

GED-SCI6-025 — Correct: A — Correct Answer: Magnetism — Explanation: Iron filings are magnetic, so a magnet can separate them from the mixture.

GED-SCI6-026 — Correct: A — Correct Answer: It modifies and packages proteins for transport. — Explanation: The Golgi apparatus modifies, sorts, and packages proteins.

GED-SCI6-027 — Correct: D — Correct Answer: Rough carpet — Explanation: The largest pulling force, 25 N, was needed on rough carpet, indicating the greatest friction.

GED-SCI6-028 — Correct: A — Correct Answer: The darker surface reflected less sunlight and became warmer. — Explanation: Dark asphalt reflected only 8% of sunlight and reached the higher temperature, 47°C.

GED-SCI6-029 — Correct: A — Correct Answer: Mutualism — Explanation: Mutualism is a relationship in which both organisms benefit.

GED-SCI6-030 — Correct: C — Correct Answer: Soap solution — Explanation: The soap solution has the highest pH, 10, making it the most basic.

GED-SCI6-031 — Correct: A,B — Correct Answer: Use the same temperature for all containers. | Repeat the test with several containers at each moisture level. — Explanation: Controlling temperature and repeating trials improves the reliability of the investigation.

GED-SCI6-032 — Correct: 4 — Tolerance: exact — Correct Answer: 4 — Explanation: $\text{Power} = 48 \text{ J} \div 12 \text{ s} = 4 \text{ watts}$.

GED-SCI6-033 — Correct: A — Correct Answer: Succession — Explanation: Succession is the gradual change in a community over time after a disturbance or in a new area.

GED-SCI6-034 — Correct: A — Correct Answer: A dominant trait passed from affected parents to some offspring — Explanation: A trait that appears in every generation and is passed from an affected parent to about half the children is consistent with dominant inheritance.